

Near-Lossless Long Context under Fixed VRAM via Critical-Span Retention

Portfolio / workshop-style draft

Lab: RTX 3090 24 GB Primary: Qwen/Qwen3-4B-Instruct-2507

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Abstract

Long-context inference is limited by KV-cache memory. Training-free eviction methods shrink the cache, but it is often unclear *which* tokens are necessary for near-full-KV quality. On a multi-seed controlled retrieval suite (Qwen3-4B, RTX 3090), we show three linked results:

1. **Structure (H1')**. Retaining critical fact tokens plus a small local radius $R^* = 1$ (with attention sinks and a recent/question window) is **necessary and sufficient** for $\varepsilon \approx 0$ single-fact recall: oracle **15/15**, anti-oracle **0/15**, full KV **15/15** (5 seeds $\times$ 3 depths @4k).
2. **Discovery gap**. Online streaming at a moderate peak budget fails under multi-seed hay not because the budget is too small, but because **query-unknown discovery is weak**: attention stream@512 is **33%**, while perfect online pin of critical $\pm R$ is **15/15** at the same peak.
3. **Closing the gap**. A **surface novelty detector** (within-prefix rarity + digit/ID-like cues; no final question) with **sticky pin packing** reaches $\sim$ **93%** multi-seed @4k and **100%** (3×3) through **40k** at stream@512—peak cache $\sim$ **1k tokens**, flat in L .

The critical-span mechanism transfers across Qwen2.5, Llama-3.2, and hybrid Gemma-4 (full layers). We do **not** claim general long-context SOTA; see Section 6.

1 Introduction

KV cache memory grows with context length L . On a fixed workstation (here: 24 GB), full-cache decode becomes the bottleneck long before “the model cannot attend.” Training-free compressors (observation-window scoring, recent-only eviction, SnapKV-style selection) reduce tokens kept, but practice often optimizes average scores rather than **guaranteeing** the spans that encode answers.

We study **near-lossless** compression on retrieval-critical prompts: quality Q should match full KV at $\varepsilon \approx 0$ under a multi-seed protocol. The systems objective is

$$L_\varepsilon = \max \{ L : Q(M, L) \geq (1 - \varepsilon) Q(\text{Full}, L) \} \quad (1)$$

under **peak** cache / VRAM constraints—not only final decode size after a full prefill.

Claim. Near-lossless training-free KV compression is better understood as **retaining critical local neighborhoods** (and discovering them online before the question exists) than as uniform thinning or posthoc score ranking alone.

2 Related work (sketch)

- **Eviction / selection**: H2O, SnapKV, pyramid and related training-free keepers.
- **Hybrid attention**: sliding-window + full layers (e.g. Gemma-4); full-layer KV still dominates compressible state.
- **Quantization** (KIVI, etc.): complementary; we use fake-int8 only for logical byte accounting.

Positioning. Mechanism + workstation L_ε , not a leaderboard chase. Novelty is a **query-unknown discovery** layer on top of H1' retention, not a new attention kernel.

3 Methods (short)

Task. Needle-in-haystack style secrets in seeded filler; depths $\{0, 0.5, 1\}$; success = exact key recall (greedy decode).

H1' keep set. $\text{sinks} \cup \text{critical} \pm R \cup \text{recent/question window}$. Kill arms: full, oracle_r1, anti_oracle.

Posthoc scorer. `seed_valley`: shared obs-window attention $\rightarrow$ seeds $\rightarrow$ contiguous valley $+ \pm R$.

Streaming. Chunked prefill; compress when cache $>$ budget; peak $\approx$ budget + chunk.

Oracle-online upper bound. Force-keep $\text{critical} \pm R$ whenever still in cache (perfect discovery).

Surface novelty (query-unknown). Per-token score from within-prefix rarity, first occurrence, and surface cues (digits, ID-like pieces). Expand peaks $\pm R$ into pin sets. **Sticky** registry: once discovered, pins stay until evicted; packing ranks pins by score under budget (sinks + recent reserved).

Default API. `prefill_auto(..., mode="stream", discovery="novelty")`.

4 Results

4.1 Figure 1 — Story in four panels

A. H1' kill (4k, 5×3 multi-seed). Full and oracle $\text{crit} \pm 1$ both **15/15**; anti-oracle **0/15**. Bare critical tokens without radius often corrupt trailing digits (H1 needs local context); $R^* = 1$ restores $\varepsilon \approx 0$. Mean $|\text{oracle}| \approx \mathbf{149}$ tokens vs full $\sim 4\text{k}$.

B. Discovery gap (stream@512). Attention valley **33%**; surface novelty **93%** (14/15); oracle pin **100%**. Peak budget is in the same class ($\sim 1\text{k}$ with chunk). Failures of valley at 512 are **detector** failures, not “stream cannot work.”

C. Long L multi-seed (3×3). Sticky novelty@512 is **9/9** at 16k, 24k, 32k, and **40k**. Non-sticky novelty degraded at 24k ($\sim 6/9$) via re-rank thrash; sticky packing fixed it. Valley@512 remains end-depth-only ($\sim 33\%$) under multi-seed hay as L grows.

D. Peak cache. Full KV scales with L ; sticky novelty stream@512 stays $\sim \mathbf{1k}$ tokens. Prior valley operating points often used $\sim 1.5\text{--}2\text{k}$ peak for long L .

4.2 Scorer tax (posthoc, question known)

Table 1: Posthoc budget tax vs oracle keep set (4k multi-seed).

Method	All-cell $B_{\min}$	Tax vs mean $ \text{oracle} $
oracle_r1	~ 149	$1.0\times$
seed_valley	192	$\sim \mathbf{1.24\times}$ global / $\sim \mathbf{1.16\times}$ mean

Near-lossless KV: critical spans → discovery gap → sticky novelty

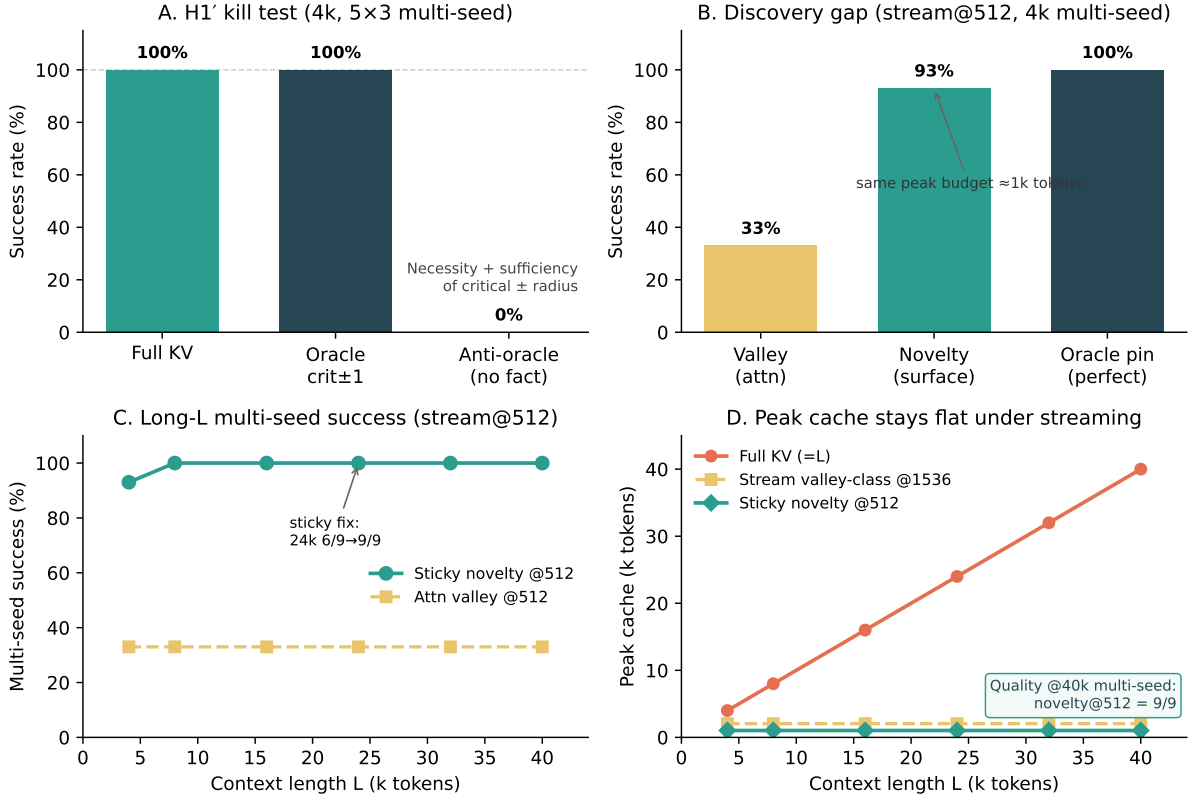


Figure 1: **Story figure.** (A) H1' kill test @4k multi-seed (5×3): full and oracle crit±1 both 15/15; anti-oracle 0/15. (B) Discovery gap at stream@512: valley 33%, surface novelty 93%, oracle pin 100%. (C) Long- L multi-seed success: sticky novelty@512 holds 9/9 through 40k; valley@512 stays end-only (~33%). (D) Peak cache tokens stay flat (~1k) under sticky novelty stream@512 while full KV grows with L .

Posthoc is near-oracle-tight; **stream** is **not**, until discovery improves.

4.3 Stress and transfer (summary)

4.4 Systems resources (peak VRAM / latency)

Mid-depth needle, seed 0. Novelty stream@512 keeps peak VRAM and decode KV essentially flat:

Peak cache under novelty stays ~1024 tokens at all three lengths.

5 Adaptive policy (systems)

`prefill.auto` applies L -based budgets + per-model floors. With default `discovery="novelty"`, single-needle stream@512 is the measured multi-seed operating point through **40k**. Attn-only discovery still needs larger floors (~1536 multi-seed @4k). Hybrid Gemma: compress **full** layers only; sliding layers keep absolute prefix bookkeeping for score-pass.

Table 2: Stress suite and cross-model transfer (summary).

Check	Result
NL facts (names/places) sticky novelty@512	15/15 multi-seed
Code needles sticky	15/15
Adversarial ID-like filler (pre-sticky)	novelty 100% vs valley 33%
Multi-needle recall_all @512	novelty 5/5 (max_new $\geq$ 96); valley 0/5
2-hop (Alice $\rightarrow$ id $\rightarrow$ password) @512	novelty 5/5 ; valley 1/5 (needs @1024 for 5/5)
3-hop + distractors @512	novelty 5/5 ; valley 0/5 (needs @1024 for 5/5)
Gemma-4 E4B hybrid novelty@512	multi-seed 9/9 @4k (valley needs $\sim$ 1024)
Qwen2.5 / Llama-3.2	H1 holds; family-specific posthoc floors

Table 3: Peak resources: full chunked prefill vs sticky novelty stream@512.

L	full VRAM	nov. VRAM	full KV	nov. KV	nov. prefill
4k	8.5 GB	8.3 GB	569 MB	72 MB	1.6 s
16k	10.6 GB	8.3 GB	2296 MB	72 MB	5.9 s
40k	14.5 GB	8.3 GB	5754 MB	72 MB	17.6 s

6 Limitations and what we do *not* claim

6.1 What we claim

- On this **controlled multi-seed retrieval suite**, $\text{critical} \pm R^*$ is necessary/sufficient for $\varepsilon \approx 0$ single-fact recall.
- Stream failures at moderate budget are primarily a **query-unknown discovery** problem (oracle-online upper bound).
- Sticky surface novelty **closes most of that gap** through **40k** multi-seed at peak cache $\sim$ 1k on the primary model, with transfer smoke to other small instruct models including hybrid Gemma-4.

6.2 What we do *not* claim

6.3 Threats to validity

- **Greedy decode** and chat templates may interact with exact-match scoring.
- **Filler distribution** is synthetic; natural corpora may change novelty baselines.
- **Hybrid models**: only full layers are compressed; sliding-window behavior is infrastructure-sensitive.
- **Selection bias toward positive arms**: we iterated methods until discovery worked; negative results (scored capsules alone, non-sticky long- L) are reported but the path is research-in-the-loop.

7 Conclusion

Near-lossless training-free KV compression on retrieval tasks is structured by **critical local neighborhoods**. When the question is known, a simple attention valley scorer approximates the oracle keep set with small tax. When the question is **not** known—as in online streaming—the binding constraint is **discovery**. Sticky surface novelty is a lightweight query-unknown detector that, on our multi-seed suite, restores high success through **40k** context at a **flat $\sim$ 1k-token peak cache**, transferring to hybrid Gemma-4 at primary-class budgets. The honest next steps are broader tasks and multi-entity packing—not larger single-needle grids alone.

Table 4: Scope boundaries (honest).

Not claimed	Why
General long-context SOTA	No full RULER / LongBench / InfiniteBench leaderboard evaluation.
All task types	Suite is retrieval / needle-class (+ limited multi-needle and NL facts). Reasoning chains, code repos, multi-doc QA largely untested.
Oracle-tight online budgets for free	Sticky novelty $\approx$ oracle quality on this suite at@512, but multi-secret packing ($\sim 80\%$ multi3) and suite-alignment remain.
Production memory stack	Fake-int8 is logical accounting only; no fused kernels or serving productization.
Novelty as universal “importance”	Exploits surface distinctness vs repetitive filler; ID-like filler / filler-like secrets can still confuse discovery.
Large models / training-free SOTA	Primary evidence is $\sim 3\text{--}4\text{B}$ instruct models on one GPU class.
Statistical finality	Multi-seed N modest (5×3 @4k; 3×3 long- L). Lab-grade, not a large meta-analysis.
$\varepsilon = 0$ beyond measured envelope	Sticky multi-seed measured through 40k ; longer L is extrapolation.
Peak VRAM = final decode KV	Streaming caps peak cache tokens ; weights and activations still dominate total VRAM.

Reproducibility

```
# Multi-seed H1 + scorer tax
python experiments/bench_paper_rigor.py --seeds 0,1,2,3,4 --ctx 4096

# Discovery gap / novelty
python experiments/bench_capsules.py --novelty --oracle-online \
  --skip-capsules --budgets 512
python experiments/bench_novelty_stress.py
python experiments/bench_novelty_longL.py \
  --ctx 16384,24576,32768,40960 --arms novelty --budgets 512

# Figure
python experiments/plot_paper_figures.py

# Transfer
python experiments/bench_transfer.py --model google/gemma-4-E4B-it
```

Primary model: Qwen/Qwen3-4B-Instruct-2507, transformers, CUDA bf16, RTX 3090.

Artifact index: results/FINDINGS.md.

Markdown twin: papers/PAPER_DRAFT.md.

Build PDF: from papers/, run powershell -File build-pdf.ps1.